

III.

$$(a) \quad \frac{\partial \left((y - X\beta)^T (y - X\beta) + \lambda \beta^T \beta \right)}{\partial \beta}$$

$$= -2X^T(y - X\beta) + 2\lambda\beta = 0$$

$$\Rightarrow X^T y - X^T X \beta - \lambda \beta = 0$$

$$\Rightarrow (X^T X + \lambda I) \beta = X^T y$$

$$\Rightarrow \hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$$

$$= (I + \lambda (X^T X)^{-1})^{-1} \hat{\beta}$$

IV

$$(b) \quad \text{LOCV} = \frac{1}{n} \text{PRESS} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_{(-i)})^2$$

$$\hat{\beta}^R = (X^T X + \lambda I_p)^{-1} X^T y$$

$$\hat{\beta}_{(-i)}^R = (X_{(-i)}^T X_{(-i)} + \lambda I_p)^{-1} X_{(-i)}^T y_{(-i)} = (X^T X + \lambda I_p - x_i x_i^T)^{-1} (X^T y - x_i y_i)$$

$$= \left\{ (X^T X + \lambda I_p)^{-1} + \frac{(X^T X + \lambda I_p)^{-1} x_i x_i^T (X^T X + \lambda I_p)^{-1}}{1 - x_i^T (X^T X + \lambda I_p)^{-1} x_i} \right\} (X^T y - x_i y_i)$$

$$= \hat{\beta}^R - (X^T X + \lambda I_p)^{-1} x_i y_i + \frac{(X^T X + \lambda I_p)^{-1} x_i x_i^T (X^T X + \lambda I_p)^{-1} (X^T y - x_i y_i)}{1 - x_i^T (X^T X + \lambda I_p)^{-1} x_i}$$

$$= \hat{\beta}^R - \frac{(X^T X + \lambda I_p)^{-1} x_i (y_i - \hat{y}_i(\lambda))}{1 - x_i^T (X^T X + \lambda I_p)^{-1} x_i}$$

$$y_i - \hat{y}_{(-i)} = y_i - x_i^T \left\{ \hat{\beta}^R - \frac{(X^T X + \lambda I_p)^{-1} x_i e_i(\lambda)}{1 - x_i^T (X^T X + \lambda I_p)^{-1} x_i} \right\}$$

2.

$$(a) \lambda \leq Q(p),$$

$$F_{\pm}(y) = (1 \pm z)F(y) + z\beta_1$$

$$= (1 \pm z)F(y) + z$$

$$= p$$

$$F(y) = \frac{p-z}{1-z} \Rightarrow y = F^{-1}\left(\frac{p-z}{1-z}\right)$$

$$\Sigma F(\lambda; T, F) = \frac{d}{dz} F^{-1}\left(\frac{p-z}{1-z}\right) \Big|_{z=0}$$

$$= \frac{p-1}{f(Q(p))}$$

$$\lambda > Q(p),$$

$$F_{\pm}(y) = (1 \pm z)F(y) + z\beta_1$$

$$= (1 \pm z)F(y)$$

$$= p$$

$$\Rightarrow F(y) = \frac{p}{1-z} \leadsto y = F^{-1}\left(\frac{p}{1-z}\right)$$

$$\Sigma F(\lambda; T, F) = \frac{d}{dz} F^{-1}\left(\frac{p}{1-z}\right) \Big|_{z=0}$$

$$= \frac{p}{f(Q(p))}$$

$$\Sigma F(\lambda; T, F) = \begin{cases} \frac{p-1}{f(Q(p))} & , \lambda \leq Q(p) \\ \frac{p}{f(Q(p))} & , \lambda > Q(p) \end{cases}$$

(b) Yes. ($\because f(x) > 0$)

$$\begin{aligned} (c) \quad V(T, F) &= \int I F(x; T, F)^2 dF(x) \\ &= p \left(\frac{p-1}{f(\theta(p))} \right)^2 + (1-p) \left(\frac{p}{f(\theta(p))} \right)^2 \\ &= \frac{p(p-1)^2 + (1-p)p^2}{f^2} \\ &= \frac{p(1-p)}{f^2} \end{aligned}$$

$$\sqrt{n} (T(F_n) - T(F)) \xrightarrow{d} N\left(0, \frac{p(1-p)}{f(\theta(p))^2}\right)$$

[3].

$$\mathbb{E}y = X\beta$$

$$\text{Var}y = \tau^2 z z^T + \Sigma$$

$$\text{Cov}(y, y) = \tau^2 z z^T$$

$$\hat{\gamma} = 0 + \tau^2 z z^T (\tau^2 z z^T + \Sigma)^{-1} (y - X\beta)$$

$$\text{let } \hat{\gamma} = a + B y$$

$$\begin{aligned}\text{Var}(\hat{\gamma}) &= B \text{Var}(y) B^T \\ &= B (\tau^2 z z^T + \Sigma) B^T\end{aligned}$$

$$\mathbb{E}(\hat{\gamma} - \gamma) = a + B X \beta$$

$$\mathbb{E}(\hat{\gamma} - \gamma) = 0$$

$$\begin{aligned}\text{TMSE}(\hat{\gamma}) &= \mathbb{E}[(\hat{\gamma} - \gamma)^T (\hat{\gamma} - \gamma)] \\ &= \text{tr}(\text{Var}(\hat{\gamma} - \gamma)) + \mathbb{E}(\hat{\gamma} - \gamma)^T \mathbb{E}(\hat{\gamma} - \gamma)\end{aligned}$$

$$\begin{aligned}\text{Var}(\hat{\gamma} - \gamma) &= \text{Var}(\hat{\gamma}) + \text{Var}(\gamma) - \text{Cov}(\hat{\gamma}, \gamma) - \text{Cov}(\gamma, \hat{\gamma}) \\ &= B (\tau^2 z z^T + \Sigma) B^T + \tau^2 I - \tau^2 B z - \tau^2 z^T B^T\end{aligned}$$

$$\begin{aligned}\text{tr}(\text{Var}(\hat{\gamma} - \gamma)) &= \text{tr}(B (\tau^2 z z^T + \Sigma) B^T) \\ &\quad + \text{tr}(\tau^2 (I - 2Bz))\end{aligned}$$

$$\text{Bias}(\hat{\gamma})^T \text{Bias}(\hat{\gamma}) = (a + B X \beta)^T (a + B X \beta)$$

$$\text{var}(\hat{\gamma} - \gamma) = \tau^4 \varepsilon^T (\tau^2 z z^T + \Sigma)^{-1} \varepsilon \\ + \tau^2 I - 2\tau^4 \varepsilon^T (\tau^2 z z^T + \Sigma)^{-1} \varepsilon$$

$$\text{tr}(\text{var}(\hat{\gamma} - \gamma)) - \text{tr}(\text{var}(\hat{\gamma} - \gamma))$$

$$= \text{tr}(B(\tau^2 z z^T + \Sigma)B^T \\ - \tau^2 2B\varepsilon + \tau^4 \varepsilon (\tau^2 z z^T + \Sigma)^{-1} \varepsilon)$$

$$= \text{tr}(B - \tau^2 z (\tau^2 z z^T + \Sigma)^{-1})^T (\tau^2 z z^T + \Sigma) (B - \tau^2 z (\tau^2 z z^T + \Sigma)^{-1})$$

$$\text{TMSE}(\hat{\gamma}) - \text{TMSE}(\hat{\gamma})$$

$$= \text{tr}(B - \tau^2 z (\tau^2 z z^T + \Sigma)^{-1})^T (\tau^2 z z^T + \Sigma) (B - \tau^2 z (\tau^2 z z^T + \Sigma)^{-1}) \\ + (a + Bx\beta)^T (a + Bx\beta) \geq 0$$

□